Overview: Reasoning and Uncertainty

In real-world reasoning, artificial intelligence systems often face uncertainty. Unlike logic-based systems, which assume that all knowledge is precise and complete, many AI problems deal with incomplete, inconsistent, or ambiguous information.

Uncertainty arises because the world is complex; data can be noisy, knowledge can be imprecise, and expert opinions may differ. Therefore, reasoning under uncertainty is a crucial part of building intelligent systems that can make sensible decisions even when all facts are not known.

When reasoning about real-world problems, we often lack perfect information. The available knowledge may be missing, inexact, or inconsistent, leading to uncertainty in conclusions. While traditional logic (such as predicate logic) can theoretically express these uncertainties, in practice it becomes cumbersome. It requires explicit predicates or complex functions to describe uncertainty, which is not intuitive or computationally efficient.

For this reason, many expert systems and AI reasoning frameworks have introduced special mechanisms to handle uncertainty. However, these mechanisms are sometimes ad hoc - that is, designed for a specific case without a strong theoretical foundation.

The goal of this topic is to understand and formalize the ways in which uncertainty can be managed logically and effectively in AI.

Sources of Uncertainty and Inexactness in Reasoning

Reasoning under uncertainty is common in AI because the knowledge used to make decisions is rarely perfect.

Some of the key sources include:

1. **Heuristic knowledge**

Heuristics are rule-of-thumb strategies used by human experts. These rules often come from experience rather than mathematical certainty, and therefore contain an element of approximation.

2. **Empirical associations**

These are relationships discovered from data or observation, which might not always hold true in all cases.

3. **Experiential reasoning**

Experts often rely on limited experiences. For example, a doctor may base a diagnosis on a small number of past cases, which might not represent all possibilities.

4. **Probabilistic knowledge**

Some knowledge is inherently probabilistic. Probabilities may be:

- **Objective** - based on statistical data (e.g., frequency counts)
- **Subjective** - based on expert judgment or belief

5. **Reproducibility**

If the same observations do not always produce the same results when repeated, the system faces uncertainty about consistency.

Dealing with Uncertainty

When reasoning under uncertainty, we must ask:

- Can the concepts used by humans be represented adequately by the system?
- Can the confidence of experts in their decisions be expressed numerically?
- Can the reasoning process be comprehensible and correct?

Important challenges include:

- **Correctness:** Do probabilities adhere to formal probability laws?
- **Relevance:** Sometimes probabilities don't sum exactly to 1, but the system still chooses the most likely outcome.
- **Long inference chains:** These often lead to extreme results (probability near 0 or 1) or meaningless mid-values (0.5).
- **Computational complexity:** Are the calculations feasible in real-time applications?

Sources of Uncertainty in AI Systems

1. **Data:** May be missing, unreliable, ambiguous, inconsistent, or subjective.
2. **Expert knowledge:** Different experts may provide conflicting opinions.
3. **Plausibility:** Experts may make "best guesses" when certain knowledge is unavailable.
4. **Causal knowledge:** Deep causal understanding is often incomplete.
5. **Observations:** May apply only to a limited domain.
6. **Representation:** The model might not capture the full complexity of reality.
7. **Inference process:**
    - **Deductive reasoning:** Produces logically correct results, but may be impractical.
    - **Inductive reasoning:** Generates new conclusions but may not always be reliable.

Uncertainty in Individual Rules

Rules in an expert system may carry their own uncertainties due to:

- **Domain errors** (incorrect assumptions)
- **Representation errors** (improperly coded rules)
- **Likelihood of evidence** (how strong each premise is)
- **Combination of evidence** (how multiple premises interact)

Uncertainty and Multiple Rules

When multiple rules are applicable:

- **Conflict resolution**: Decide which rule to prioritize.
    - May depend on **explicit priorities** (set by experts) or **implicit ones** (derived from rule properties).
- **Compatibility**: Rules may contradict each other.

- **Subsumption**: One rule may be a more general version of another.
- **Redundancy**: Some rules may overlap unnecessarily.
- **Data fusion**: Integrate data from multiple sources to make a consistent decision.

Probability Theory

1. **Basics**

Probability theory provides a mathematical framework for handling uncertain information.
Let the sample space be:
$X=\{x1,x2,...,xn\}$
Each xi represents a possible event.
Key properties:

- P(xi) lies between 0 and 1.
- The total probability of all possible events = 1.
- For mutually exclusive events, the probability of either occurring is the sum of their probabilities.

Experimental probability
Based on observation or frequency of occurrence.
Subjective probability
Based on expert belief or experience.

2. **Compound Probabilities**

When dealing with independent events (events that do not affect each other):
$P(A,B) = P(A) \times P(B)$
Example: Rolling two dice
The probability of both showing 6 is:
$P(6,6)=1/6\times1/6=1/36$

3. **Conditional Probabilities**

When one event depends on another:
$P(A \mid B)=P(A \text{ and } B) / P(B)$
Example: The probability that it rains given that the sky is cloudy.

Advantages and Problems of Probability
Advantages

- Solid mathematical foundation
- Realistic for frequency-based data

Problems

- May not reflect future situations accurately
- Requires precise numerical values
- Must assign probabilities even when information is missing
- Difficult to handle interdependent evidence

- Computationally demanding for large systems

Bayesian Approaches
Bayesian reasoning calculates the probability of a cause given an observed effect.
For example, in medical diagnosis:

- You observe a symptom.
- You want to find the probability that it was caused by disease D.

$P(D \mid S) = P(S \mid D) \times P(D) / P(S)$
This is the foundation of **Bayesian networks**, which are used widely in diagnostics, machine learning, and decision systems.

Advantages:

- Strong theoretical foundation
- Clear decision semantics

Problems:

- Requires large datasets
- Assumes independence among evidences (often false)
- Hard to interpret for end users
- Computationally heavy

Certainty Factors, Belief and Disbelief
Certainty factors (CF) represent the degree of belief in a hypothesis H given evidence E.
$CF(H,E) = \text{Belief}(H \mid E) - \text{Disbelief}(H \mid E)$

Originally introduced in the **MYCIN** expert system for medical diagnosis, CFs allow reasoning even when evidence is partial.
Advantages:

- Easy to implement
- Captures human-like reasoning
- Requires no statistical data

Problems:

- Ad hoc origin (formal foundation added later through Dempster-Shafer theory)
- Poor handling of dependent evidence
- Not suitable for long inference chains

Dempster-Shafer Theory and Evidential Reasoning
The Dempster-Shafer (DS) theory extends probability theory by representing uncertainty with intervals of belief rather than single-point values.
It distinguishes between:

- **Belief** (support for a hypothesis)

- **Disbelief** (support against it)
- **Ignorance** (lack of knowledge)

DS theory combines multiple sources of evidence using mass functions, emphasizing overlapping (consistent) evidence and discounting conflicting information.
Evidential reasoning extends DS theory to handle imprecise and possibly inaccurate knowledge.

Advantages:
- Rigorous foundation
- Can model ignorance
- Represents uncertainty more flexibly

Problems:
- Complex computation
- Sometimes counterintuitive results
- Difficult to interpret mass probabilities

Revision Questions
1. What are the main sources of uncertainty in AI reasoning?
2. Explain how probability theory deals with uncertain information.
3. Compare Bayesian reasoning with certainty factors.
4. What are the advantages and disadvantages of Dempster-Shafer theory?
5. Why is reasoning under uncertainty important in AI?

Exercises
What is statistical classification?
Discuss the Bayesian classification technique, including Bayes' Rule for single and multiple events.
Using a suitable example, demonstrate how Bayes' theorem can be applied to solve a classification problem.